

Table 1: Collision cost, cost_limit=0: SafeDreamer (100k steps) vs MACPO (5M steps).

Algorithm	Scenario	Score $\uparrow$	Cost $\downarrow$	Winrate $\uparrow$	Steps	Cost Limit
MACPO	2s_vs_1sc	20.16 $\pm$ 0.09	0.00 $\pm$ 0.00	0.99 $\pm$ 0.01	5M	0
Safe Dreamers	2s_vs_1sc	19.86 $\pm$ 0.22	0.00 $\pm$ 0.00	0.95 $\pm$ 0.03	100k	0
MACPO	3m	19.85 $\pm$ 0.11	0.00 $\pm$ 0.00	0.99 $\pm$ 0.01	5M	0
Safe Dreamers	3m	19.03 $\pm$ 0.91	0.02 $\pm$ 0.02	0.92 $\pm$ 0.07	100k	0
MACPO	2s3z	18.78 $\pm$ 0.04	0.02 $\pm$ 0.00	0.80 $\pm$ 0.01	5M	0
Safe Dreamers	2s3z	18.72 $\pm$ 0.44	0.16 $\pm$ 0.11	0.79 $\pm$ 0.07	100k	0
MACPO	3s_vs_3z	19.55 $\pm$ 0.35	0.00 $\pm$ 0.00	0.94 $\pm$ 0.05	5M	0
Safe Dreamers	3s_vs_3z	19.89 $\pm$ 0.05	0.02 $\pm$ 0.03	0.97 $\pm$ 0.01	100k	0
MACPO	3s_vs_4z	9.18 $\pm$ 0.86	0.00 $\pm$ 0.00	0.00 $\pm$ 0.00	5M	0
Safe Dreamers	3s_vs_4z	15.49 $\pm$ 3.71	0.26 $\pm$ 0.23	0.06 $\pm$ 0.09	100k	0
MACPO	3s_vs_5z	7.16 $\pm$ 0.76	0.00 $\pm$ 0.00	0.00 $\pm$ 0.00	5M	0
Safe Dreamers	3s_vs_5z	13.72 $\pm$ 2.97	0.08 $\pm$ 0.07	0.01 $\pm$ 0.03	100k	0
MACPO	8m	19.77 $\pm$ 0.07	0.32 $\pm$ 0.05	0.97 $\pm$ 0.01	5M	0
Safe Dreamers	8m	17.49 $\pm$ 0.43	0.58 $\pm$ 0.06	0.71 $\pm$ 0.05	100k	0
MACPO	MMM	11.13 $\pm$ 0.17	0.96 $\pm$ 0.10	0.00 $\pm$ 0.00	5M	0
Safe Dreamers	MMM	15.12 $\pm$ 0.17	1.20 $\pm$ 0.52	0.39 $\pm$ 0.03	100k	0
MACPO	3s5z_vs_3s6z	9.52 $\pm$ nan	0.11 $\pm$ nan	0.00 $\pm$ nan	5M	0
Safe Dreamers	3s5z_vs_3s6z	10.65 $\pm$ 0.82	0.86 $\pm$ 0.08	0.01 $\pm$ 0.01	100k	0
MACPO	bane_vs_bane	—	—	—	5M	0
Safe Dreamers	bane_vs_bane	18.20 $\pm$ 0.59	0.15 $\pm$ 0.13	0.57 $\pm$ 0.10	100k	0